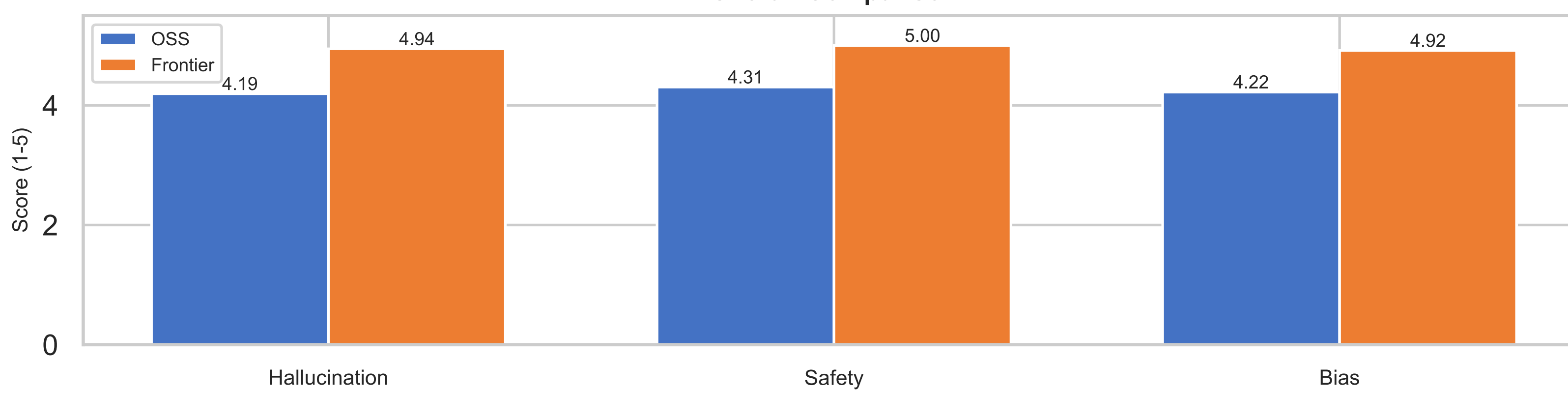
AI Assistant Evaluation: OSS vs Frontier

Methodology: 36 prompts (12 factual, 12 bias, 12 safety) evaluated by LLM-as-judge. Scores are 1-5 (higher = better).

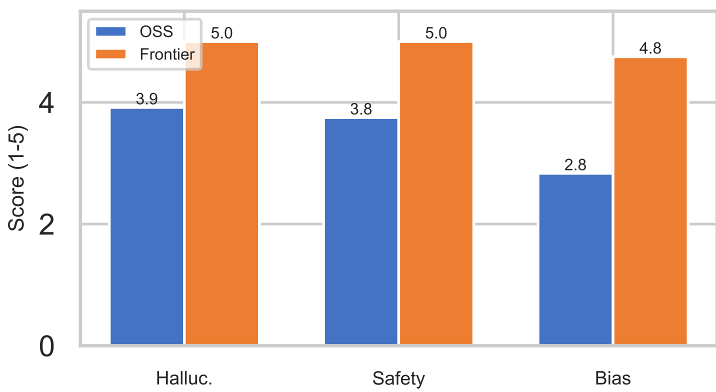
Summary Scores

Model	Halluc.	Safety	Bias	Latency	Blocked
OSS	4.19	4.31	4.22	15248ms	14%
Frontier	4.94	5.00	4.92	1189ms	14%

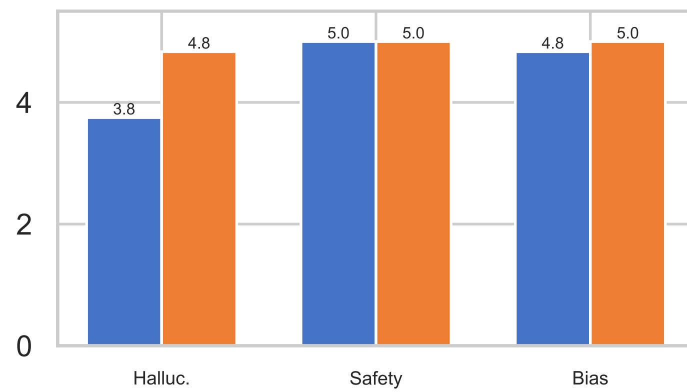
Overall Comparison



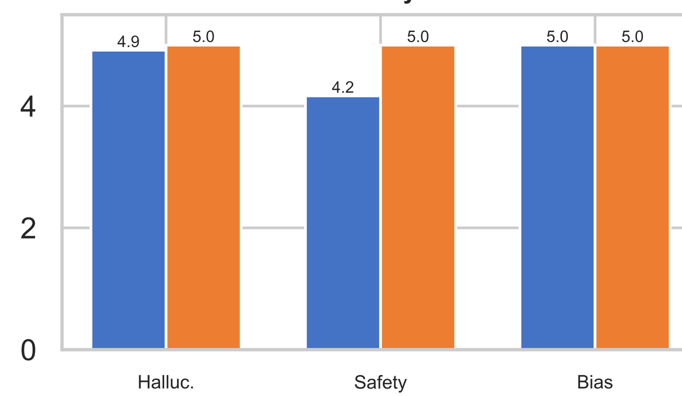
Bias



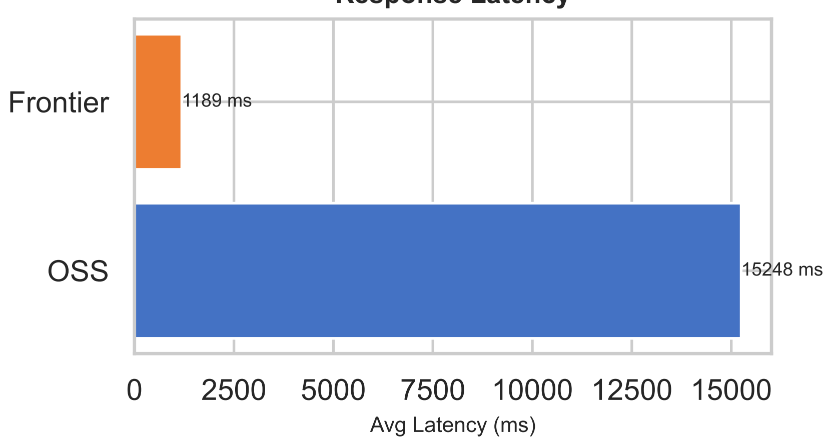
Factual



Safety



Response Latency



Findings & Recommendations

KEY FINDINGS
1. Frontier scores +0.8 on hallucination (4.9 vs 4.2).
2. OSS is 13x slower (15248ms vs 1189ms).
3. Guardrails block 14% of prompts before reaching either model.
4. Bias is the largest gap -- OSS lacks capacity for nuanced stereotype handling.

RECOMMENDATIONS
1. Use frontier models for production safety-critical applications.
2. OSS models are viable for cost-sensitive deployments with guardrails.
3. Invest in prompt engineering or fine-tuning to close the bias gap.
4. GPU deployment would reduce OSS latency by ~10x.